AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

farmX: Smart Farming App

Team ID: *GEN26-A0H-I045*

Team Name: *Sneakers*

Domain 5: *Food Security, Agriculture & Rural Transformation*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

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Presentation Outline

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Problem Statement

Small and medium-scale farmers need accessible support to identify crop disease, plan irrigation, sell produce and protect their fields. Scattered information and language barriers make these decisions harder.

farmX brings crop care, weather, water planning, trade and field monitoring into one app, with help in the farmer's local language.

Problem Understanding

- **Who:** Small and medium-scale farmers who need timely, understandable advice and access to nearby and distant buyers.
- **Why:** Uncertain crop symptoms, changing weather, limited irrigation guidance, delivery barriers and difficulty using unfamiliar apps.
- **Consequences:** Delayed crop care, avoidable water use, difficult selling decisions and animal intrusion. Electric fences can also endanger animals.

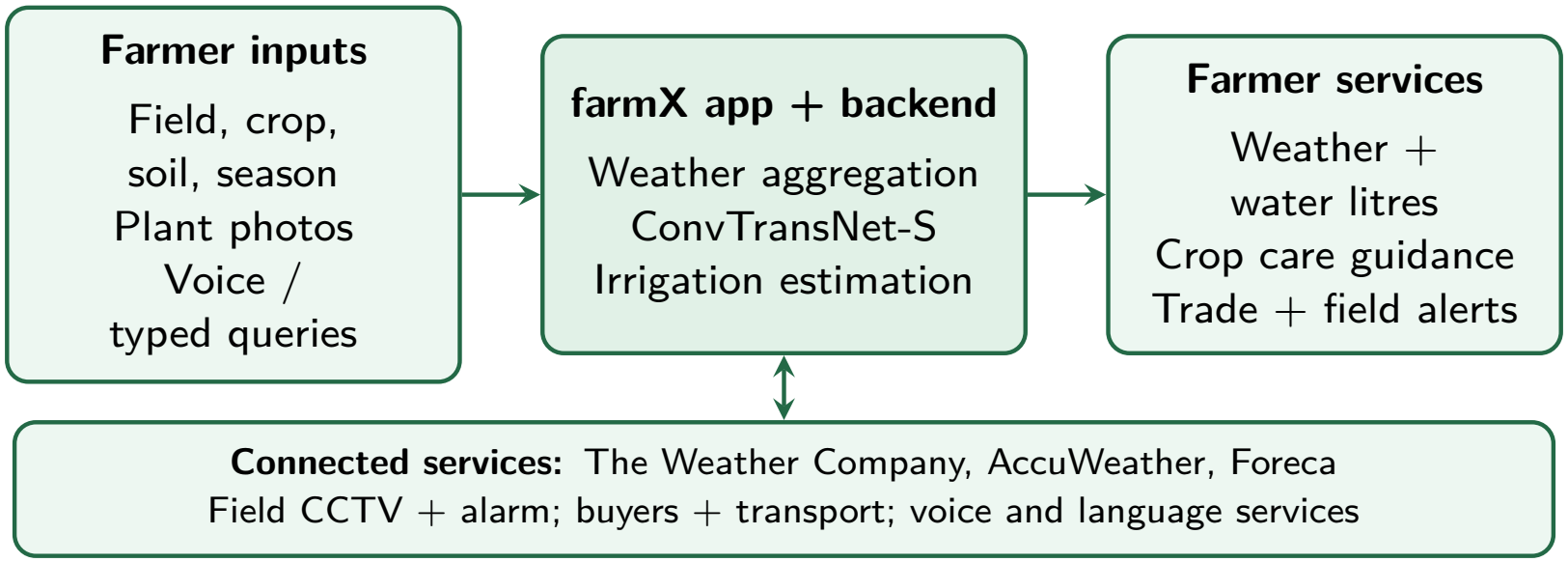
Existing Gaps & Challenges

- **Disconnected decisions:** Crop diagnosis, weather, irrigation, selling and field monitoring need to work together in the farmer's daily routine.
- **Accuracy in real fields:** Forecasts need local evaluation. Disease photos vary with lighting and background, and broader pest recognition needs training.
- **Access and adoption:** Local-language support, reliable connectivity, affordable CCTV and dependable produce delivery are essential.

Our Proposed Solution

- **Weather report:** Combine The Weather Company, AccuWeather and Foreca forecasts, aiming for more reliable local guidance.
- **Crop Doctor:** Use ConvTransNet-S for leaf disease symptoms, with further training planned for insect and pest damage.
- **Irrigation calculator:** Use field dimensions, crop, sowing season, soil and app weather to estimate water required in litres.
- **Online market:** Sell to nearby and long-distance buyers, with delivery through a contracted transport service.
- **Field monitoring:** Live CCTV, an animal-intrusion alarm and farmer alerts support crop protection and reduce reliance on electric fences.
- **AI assistant:** Answer queries and teach app use in the local language through voice chat and typed chat.

Technical Approach



Weather updates feed the irrigation estimate. Animal detection triggers a field alarm and an app alert.

Proposed Tech Stack

Frontend: Flutter

Backend: Python / FastAPI

Database: PostgreSQL

AI / APIs: ConvTransNet-S, weather, voice

Innovation & Uniqueness

- **What is new or different?** farmX combines crop diagnosis, weather, irrigation estimates, produce sales, field monitoring and AI help in one app.
- **Why does this approach suit farmers?** Watering estimates use the farmer's field, crop, soil, season and weather. Local-language voice and text make guidance easier to use.
- **What is the key differentiator?** Weather updates also inform irrigation, produce sales include partner delivery, and animal detection triggers alarms and farmer alerts to reduce reliance on electric fences.

Feasibility & Scalability

- **Feasibility:** Begin with Crop Doctor, weather, irrigation and the voice/text assistant. Pilot with farmers, then integrate CCTV, alarms, the marketplace and transport operations.
- **Scalability:** Reuse the backend across crops and regions. Add validated disease classes, local languages and delivery partners in stages.
- **Risks and checks:** Validate field diagnosis, insect/pest training, forecast accuracy and animal alerts. Plan for API licences, connectivity, hardware upkeep and transport costs.

Cost for Product (Optional)

- **Build:** Student development using existing laptops and open-source tools. Budget separately for app integration and field testing.
- **Operate:** Recurring cloud hosting, AI inference, database, weather-data licences and voice services. Fees depend on pilot usage.
- **Field kit:** CCTV camera, alarm, power and connectivity per field. Final cost requires provider quotations and the number of pilot farms.

Revenue model: 20% of each sale: 10% for contracted transport and 10% for farmX.

On a Rs. 1,000 sale: farmer Rs. 800, transport Rs. 100, farmX Rs. 100.

Planning model. Total launch budget awaits quotations and pilot scope.

Use Cases

- **1. Spot crop disease:** A farmer uploads a leaf photo to Crop Doctor and hears the result and guidance in the local language. Voice or typed queries provide further help.
- **2. Plan watering:** A farmer enters field dimensions, crop, sowing season and soil. The app combines these with its weather report to estimate water required in litres and avoid unnecessary watering.
- **3. Sell and monitor remotely:** A farmer sells produce to a local or distant buyer with partner delivery. Live field CCTV also sends an animal intrusion alert while the field alarm sounds.

Impacts and Benefits

- **For farmers:** Earlier crop-care decisions, clearer watering guidance, access to buyers and delivery, and app help through local-language voice and typed chat.
- **For farms and animals:** Aim to reduce avoidable water use, support crop health and reduce animal deaths from electric fences by using intrusion alarms and farmer alerts.
- **Measure in the pilot:** Field diagnosis accuracy, forecast error, water used, completed sales, farmer usability and animal-alert false alarms. These benefits require field validation.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 2	Target 2.4	Zero Hunger	Crop care and weather guidance support resilient, sustainable farming.
SDG 6	Target 6.4	Clean Water and Sanitation	Weather-aware irrigation estimates support efficient water use.

Intended contributions. Sources: UN SDG 2, Target 2.4; UN SDG 6, Target 6.4.

Individual Member Contribution

Member	Contribution or responsibility
S. Kaveen	**Team Leader:** Team coordination, research, app design, testing and presentation.
Alen Jomon	Weather report and irrigation calculator.
Abijith raja S	Crop Doctor and AI assistant.
Iniyan J	Online market, delivery and CCTV alerts.

Yukti ID

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